relaying Retention time, Dead time;

and Retention rate (for some molecule A)

A = Compound

$(t_R)_A$ = Retention Time

t_M = Dead time

Equation for Retention Factor

$$\frac{(t_R)_A - t_M}{t_M}$$

$$Area_{ST} = \frac{Area_x}{FC_x} \cdot C_{ST}$$

x = Compound

ST = Internal Standard

F = Retention Factor

Vector Function relaying Retention Factor, Con

$Concentration_{ST} = C_{ST}$

$Concentration\ x = C_x$

$$Area_x = F\left(\frac{Area_{ST}}{C_{ST}}\right) C_x$$

$$\left\langle F\left(\frac{Area_{ST}}{C_{ST}}\right)C_x, \left(\frac{Area_x}{FC_x}\right)C_{ST}, \left(\frac{Area_x}{C_x}\right)\left(\frac{C_{ST}}{A_{ST}}\right)\right\rangle$$

$$\langle x(u,v),\ y(u,v),\ z(u,v)\rangle$$

$$\frac{(v_R)_A - v_M}{v_M}$$

Unit vector

$$v_M \hat{i} + \left(\frac{(v_R)_A - v_M}{v_M}\right)\hat{j}$$

$$\sqrt{(v_M)^2 + \left(\frac{(v_R)_A - v_M}{v_M}\right)^2} \rightarrow \sqrt{(v_M)^2 + \left[\frac{(v_R)_A^2 - 2(v_R)_A v_M + v_M^2}{v_M^2}\right]}$$

$$\rightarrow \sqrt{(v_M)^2 + \left[\frac{(v_R)_A^2}{v_M^2} - \frac{2(v_R)_A \cancel{v_M}}{\cancel{v_M^2}\, v_M} + \frac{\cancel{v_M^2}}{\cancel{v_M^2}}\right]}$$

$$\rightarrow \sqrt{(v_M)^2 + \frac{(v_R)_A^2}{v_M^2} - \frac{2(v_R)_A}{v_M}}$$

$$\frac{v_M \hat{i} + \left(\frac{(v_R)_A - v_M}{v_M}\right)\hat{j}}{\sqrt{(v_M)^2 + \frac{(v_R)_A^2}{v_M^2} - \frac{2(v_R)_A}{v_M}}}$$